

Audio file

[6.5-6.7 LAD, Lymphoma, Hodgkin Lymphoma.mp4](#)

Transcript

Welcome back. Now that we've completed our discussion of leukemias, we can now address lymph nodes. And the first topic that we're going to discuss is lymphadenopathy. Lymphadenopathy refers to enlarged lymph nodes. And it's important to note that lymphadenopathy can be painful or painless. Painful lymphadenopathy is usually seen when there's acute infection and the lymph node is draining that region of infection. For example, let's say that the patient has a strep throat. The lymph nodes in the neck will then be enlarged and they will be painful. So that would be called painful lymphadenopathy. At the same time, it is also possible to develop a painless lymphadenopathy, and the differential diagnosis for painless lymphadenopathy includes chronic inflammation. So, for example, autoimmune diseases often have painless lymphadenopathy associated with them. Metastatic carcinoma, if carcinoma spreads to a local lymph node, it's usually painless enlargement of the lymph node that occurs. For example, if a woman has breast cancer and it spreads to the axillary lymph nodes, the axillary lymph nodes become enlarged and that's usually painless. And the third circumstance in which we can get painless lymphadenopathy is lymphoma. So when we think about painless lymphadenopathy, we really have three major differential diagnoses in our mind. Chronic inflammation, metastatic carcinoma, or lymphoma. We're going to cover lymphoma in much more detail in just a few slides. However, when we have a reactive lymphadenopathy, meaning that the lymph node is big due to some chronic inflammation or reactive process, it is important to note that there is going to be hyperplasia of differing regions of the lymph node depending on the underlying etiology. For example, if the patient has rheumatoid arthritis, it's classic for there to be expansion of the follicles, and it's this expansion of the follicles that would lead to enlargement of a lymph node. Similarly, in early HIV, there's also expansion of the follicles, and again, it's expansion of the follicles that results in the generalized lymphadenopathy that's often seen in early HIV. Now, recall that HIV infects CD4 positive cells. And of course, the most important and classic infection would be of the CD4 positive T cells. However, the cells in the follicle, the follicular dendritic cells, they also are CD4 positive. And that's why you see this follicular hyperplasia in early HIV. The paracortex is the region where the T cells live, and so it's pretty straightforward for us to understand that with viral infection, the paracortex would become expanded, and it's expansion of the paracortex that leads to lymphadenopathy. The classic example would be

infectious mononucleosis. Recall that in infectious mononucleosis, an EBV infection results in enlargement of the lymph nodes, and that's going to be due to paracortical expansion of the lymph node. And then finally, we can see enlargement of the lymph node due to expansion of the region where the histiocytes live. And of course, the histiocytes live deep in the sinus in the medulla of the lymph node. And this is classically seen when there is a lymph node that is draining a region of cancer. Let's say that a woman has breast cancer and there are some lymph nodes that are draining that region. Well, it would be classic to see sinus histiocytosis in those lymph nodes. Now, of course, the cancer hasn't gone to the lymph node in this case, but this lymph node is simply becoming reactive because it's draining a region of cancer. And so that's another classic cause of lymphadenopathy. Here's a picture of a lymph node just to remind ourselves of the normal architecture. Up here would be the cortex. Underneath the cortex is the paracortex, and in this region is the medulla. Now, of course, the cortex is where the B cells live. The paracortex is where the T-cells live. And the medulla-- can you see these lighter pink areas here? This is where the sinus histiocytes live. And so these are the three major regions of the lymph node. Of course, we'd see follicular expansion in the cortex with rheumatoid arthritis and early HIV. We'd see expansion of the paracortex with viral infections. And again, we would see expansion of the medullary region when we had a lymph node that was draining a region of cancer. Now, I put this here in your text and in this discussion because it's often that examiners like to ask about which region of the lymph node is expanded based on a particular etiology. So that's a relatively short section, and now we're gonna move into lymphoma. And basically, the very first principle that I'd like to establish is that lymphoma is a neoplastic proliferation of lymphoid cells that forms a mass. Lymph, i.e. lymphoid cells, oma, that forms a mass. And that's what gives it its name of lymphoma. Now lymphoma can arise within the lymph node, and that makes sense because that's where lymphocytes normally live, But it can also arise within extranodal tissues outside of the lymph node. When we think about lymphoma generally, we should remind ourselves that lymphoma can be divided into non-Hodgkin's lymphoma, which is the more common type, and Hodgkin's lymphoma, which is still relatively common. We will talk in more detail about Hodgkin's lymphoma a little later on. And so at this point, what I'm going to do is simply focus on the non-Hodgkin's lymphomas. When we think about the non-Hodgkin's lymphoma, we can classify them based on several parameters. These include cell type, for example, B-cell versus T-cell. Cell size, for example, small cells or intermediate cells or large cells. Pattern of growth, for example, do the cells grow in a diffuse sheet, or do they form follicles, just as an example. Expression of surface markers, in particular, which would help us to determine whether the cells are B-cells or T-cells. and also cytogenetic translocations. Now, classically, when you think about non-Hodgkin's lymphoma, there have been multiple classification schemes, and I don't want to go into the nitty-gritty in

details, but I'll just simply tell you that I think about Hodgkin's lymphoma in a rather simplistic manner. It's probably not the most accurate way if you were a hematopathologist or a hematologist, but I do think that it is an easy way to understand a very difficult concept. And so what we're going to do is I'm going to sort of base my division of non-Hodgkin's lymphoma on cell size. Now, in our discussion, we're going to be focusing on the B-cell types of non-Hodgkin's lymphoma, which are the vast majority of non-Hodgkin's lymphoma. So we're going to base it on cell size, and part of that discussion will also be based on pattern of growth, but this will become clear just in a moment. So let's go to a whiteboard and remind ourselves that really, when you think about cell size, lymphocytes, when they're neoplastic, can be small, can be intermediate, or can be large. And when the lymphocytes are small, we would think about those as well differentiated. In other words, small lymphocytes that are neoplastic are well differentiated. They look just like very similar to normal lymphocytes. So they're going to do, on occasion, when they grow, they're going to grow in the patterns in which normal lymphocytes grow. Let's go back and remind ourselves of the normal lymph node. Remember that we're focusing on B cells, and so the normal B area of the lymph node is the cortex. And in the cortex, you have something called the follicle, that's the follicle. And around the follicle is the mantle, it's a dark area that surrounds the follicle, this is called the mantle. And then, outside of the mantle, on occasion, not always, but on occasion, we'll actually have generation of something called the margin. So these are the normal zones of the B area of the lymph node. follicle, mantle, and margin. And when we think about cells that are small, that are neoplastic, they basically produce structures that are very similar to the normal lymph node. So we can have a follicular lymphoma that's going to produce a bunch of follicle-like structures. We can have expansion of the mantle zone by neoplastic cells, which is called a mantle cell lymphoma. Or we can have expansion of the marginal zone by neoplastic cells, which is called a marginal zone lymphoma. And these are the three key lymphomas that arise from small neoplastic B cells. There is a fourth one, which I'm not going to discuss here, which we've already discussed, and that's small lymphocytic lymphoma. And recall that that is when CLL involves the lymph nodes. But basically here, I'm going to focus on follicular lymphoma, mantle cell lymphoma, and marginal zone lymphoma. Follicular lymphoma is a neoplastic proliferation of small B cells. Those B cells would be CD20 positive. CD20 is a key marker for B cells. And these small B-cells are going to make follicle-like nodules, hence the name follicular lymphoma. Clinically, it presents in late adulthood and usually with painless lymphadenopathy. Now, lymphadenopathy refers to enlargement of the lymph nodes, and these patients will have generalized lymphadenopathy throughout the body that is painless. Here is a picture of follicular lymphoma, and what we can appreciate here is that we're dealing with a lymph node, and we see that the lymph node basically contains dozens of these follicles. So these lymphocytes are making follicle-like structures,

hence the term follicular lymphoma. One of the important findings in follicular lymphoma is that it is driven by a translocation of 14, 18. Important to remember, and very high yield, that what's happening here is that if this is 14, normally in a B cell, on chromosome 14 at this particular locus is immunoglobulin heavy chain. And what translocates with immunoglobulin heavy chain in follicular lymphoma is a gene called BCL2. BCL2 happens to be in chromosome 18. And so this 14, 18 translocation pushes BCL2 into the position where immunoglobulin heavy chain was once sitting. Now immunoglobulin heavy chain gene is a highly active gene in B cells. And so BCL2, when it gets translocated here, becomes highly active and becomes overexpressed. How does that create a follicular lymphoma? Let's recall that BCL-2, its role is to stabilize the mitochondrial membrane. And by stabilizing the mitochondrial membrane, it prohibits cytochrome C from leaking from the mitochondria into the cytoplasm, and that blocks apoptosis. Most cells in the human body, they express BCL-2 because you don't want apoptosis to occur, and you want to hide away cytochrome C. However, in the follicle of the lymph node, that's the one place, that's the particular place in the human body where you actually want apoptosis to occur. Why? Well, what's normally going on in the follicle? In the follicle, you get somatic hypermutation of the B cells. The B cells come in, and they are challenged by antigen, and they undergo somatic hypermutation to see whether they can bind or produce antibodies that are going to bind against that antigen. If they pass somatic hypermutation, they then go from the germinal center, and they can then participate in inflammation. However, if they fail somatic hypermutation, which the vast majority of B cells do, then they undergo apoptosis. And when they undergo apoptosis, that allows them to die. Now, imagine if there's a translocation of the B cell so that BCL2 is overexpressed. then those cells that should have died in the follicle actually won't be able to die. And that's then going to result in a follicular lymphoma. So because we need apoptosis in the follicle, and we aren't able to undergo apoptosis due to the high BCL-2, a follicular lymphoma will develop. And that's what I'm highlighting here in your slide. where I say that it's driven by a 1418 translocation, and that involves BCL-2 on chromosome 18, which translocates to the immunoglobulin heavy chain locus on chromosome 14. Again, overexpression of BCL-2 then inhibits apoptosis. These patients are relatively asymptomatic, and they live for a long period of time with this disease. Treatment is usually reserved for patients who are symptomatic, and one way by which we could treat is to give low-dose chemotherapy. Another way would be to give a drug called rituximab, which is an anti-CD20 antibody. So this is another way by which we could treat follicular lymphoma. The complication of follicular lymphoma is that it can progress to a diffuse large B-cell lymphoma. So over time, it could actually pick up additional mutations, resulting in a diffuse large B-cell lymphoma, and this would present as an enlarging lymph node. Now, recall that patients with follicular lymphoma, they have generalized lymphadenopathy, where they have multiple lymph nodes in their body that are

enlarged. If all of a sudden, one of those lymph nodes begins to become further enlarged, begins to grow more, then we think about progression to diffuse large B-cell lymphoma. It is very important for the purposes of both clinics as well as board exams to distinguish follicular hyperplasia from follicular lymphoma. Now, what do I mean by that? If a patient gets an infection, it is possible that they could get hyperplasia of the follicles just as a result of that infection. And so we need to, when we see a bunch of follicles within a lymph node, we need to be able to distinguish Is this due to follicular lymphoma, which is neoplastic, or is this simply a reaction to an infection? Now, how could we do that? Well, number one, when we have follicular lymphoma, there'll be disruption of the lymph node architecture. What do I mean by that? Well, I'm going to take you back a couple slides to this picture, which I have drawn you here, and let's look at this lymph node again. When we look at this lymph node, what we see is that there are follicles throughout the entire lymph node, and we don't see any other structures. So these follicles have basically taken over or disrupted the normal lymph node architecture, and that's an indication that we're dealing with follicular lymphoma. If this was a follicular hyperplasia in response to an infection, We would see expansion of the follicles, but all of the follicles would be present in the cortex where the follicles normally live. The fact that these follicles are destroying the lymph node and going deep into the lymph node is an indication that we may be dealing with a follicular lymphoma. The second important feature concerning the distinction between hyperplasia and lymphoma is a lack of tingible body macrophages in the germinal center. This image here is meant to highlight that. Now let's remind ourselves that when we have infection, then we're going to get a follicular hyperplasia, and that follicular hyperplasia occurs because B cells are coming into the follicle. They are undergoing somatic hypermutation in the follicle. And as a result of somatic hypermutation, many of them are dying. Now, if the B cells in the follicle are truly undergoing somatic hypermutation and dying by apoptosis, then recall that the response to apoptosis at the very end is that the macrophages come in and they clean up the dead cells. And so if we have a follicle that is reacting to infection, we should see a ton of macrophages present within the germinal center. And that's what these little white spaces here represent within this germinal center. So why don't I take you back a step and show you that this is the germinal center. This darker blue area here is actually the mantle surrounding the germinal center. And within this germinal center, we see these white spaces which indicate that macrophages are present and are eating the debris of apoptotic cells. These macrophages are also called tingible body macrophages, and they indicate, again, that apoptosis is actively occurring within the germinal center. Now, let's contrast that against follicular lymphoma. Here's the follicle of follicular lymphoma. Do you see any tingible body macrophages in that follicle? The answer is no. We don't have those nice tingible body macrophages in the center of that follicle, and that's because these cells are not undergoing apoptosis. They have undergone a 14-18 translocation, BCL-2 is

overexpressed, and the overexpression of BCL-2 is actually blocking apoptosis in these cells. And that's an important point concerning follicular lymphoma. A third important point concerning follicular lymphoma is the expression of BCL-2 in the follicles. The vast majority of cells in the human body, they express BCL-2, and that's because you don't want those cells to undergo apoptosis. However, in the follicle, in a normal follicle, in you and I, there would be no expression of BCL-2 because you want to turn BCL-2 off to allow for apoptosis. But if there is expression of BCL-2, that would indicate that we're dealing with a follicular lymphoma. The final way by which we can make this distinction is monoclonality. Recall that if we have a reactive proliferation of lymphocytes, and this goes back to our discussions of neoplasia, if you have a reactive proliferation of lymphocytes, it would be polyclonal. And if we have a neoplastic proliferation of lymphocytes, it would be monoclonal. And of course, clonality is gauged in B cells by looking at the ratio of kappa to lambda light chain. In a polyclonal reactive proliferation, the ratio would be three to one, whereas in neoplasia, the ratio would be disrupted. For example, classically on an examination, 20 to one or 25 to one. So that is follicular lymphoma. Now we come to mantle cell lymphoma. And mantle cell lymphoma is when you have a neoplastic proliferation of small B cells, obviously CD20 positive, that expand the mantle zone. This clinically also presents in late adulthood. It also presents with generalized painless lymphadenopathy. Now, when you have this type of tumor, one of the classic descriptions on examination is that the examiners could tell you that there is a neoplastic proliferation of cells that expands the region immediately adjacent to the follicle. If you have a tumor that expands the region immediately adjacent to the follicle, that region is obviously the mantle, and so that would be a mantle cell lymphoma. So that's a nice way that examiners might try to describe the tumor. This tumor is driven by an 1114 translocation. Well, what's on 14? Immunoglobulin heavy chain. What gets translocated to that spot? Cyclin D. That's the key gene that's going to drive the production of mantle cell lymphoma. Now, recall that cyclin D is a cell cycle regulator. It helps the cell to go from G1 to S phase, and the way that it does it is by phosphorylating proteins. So a high level of cyclin D would actually phosphorylate the regulators allowing the cell to go from G1 to S phase. And this is important for examination purposes. You need to know that cyclin D causes the cell to go from the G1 to S phase. And this is a key translocation in mantle cell lymphoma. The third lymphoma that would produce a structure similar to the normal lymph node is a marginal zone lymphoma. So what is the marginal zone? The marginal zone is that region which is outside of the mantle. Here's a lymph node again. This is the follicle. Outside of the follicle would be the mantle, and outside of the mantle would be the margin. And so we could develop a tumor that expands the marginal zone, and that's called a marginal zone lymphoma. Again, it's usually due to neoplastic B cells that are going to be CD20 positive. Important to know that this type of lymphoma is associated with chronic inflammatory states. This kind of goes back a

little bit to normal immunology, because remember that most lymph nodes don't have a marginal zone. The marginal zone is actually formed by when there is activation of the cells through the germinal center. And so when cells go into the germinal center and get activated because of, for example, chronic inflammation, that's when the margin is produced. And so the marginal zone would be classically seen, I should say, when there's chronic inflammatory states. So what are some associated disorders? In Hashimoto's thyroiditis, patients can develop a marginal zone lymphoma as a complication years later. In Sjogren's syndrome, as I've already described in the immunology lectures, we can develop a complication of marginal zone lymphoma, and the idea here would be that the patient presents with unilateral enlargement of the parotid years into her Sjogren's syndrome. And then another classic one is H. pylori gastritis. Now, in H. pylori gastritis, what happens is that you get a marginal zone lymphoma in mucosal sites, and that's also called a maltoma. And again, the gastric maltoma in the mucosa of the stomach can possibly regress if the H. pylori is treated. And so maltoma, mucosa-associated lymphoid tissue-oma, mass, is one of the subtypes of marginal zone lymphoma. Now that concludes our discussion of the small B-cell lymphomas that mimic normal structure of the lymph node. And so we're gonna move now to a tumor in which we have intermediate-sized B-cells. Of course, they're CD20 positive because they're B-cells, and this is called Burkitt lymphoma. So Burkitt lymphoma, intermediate-sized B-cells that are CD20 positive. There is an association with EBV. Classically, this tumor presents as an extranodal mass in a child or young adult. There are two forms. It's the African form, which produces a jaw mass, and there's a sporadic form, which involves the abdomen. One of the ways by which examiners like to go after this is they show you a picture like this, and when they show you a picture like this, they are trying to have you recognize that this is an example of Burkitt's lymphoma of the jaw, and then you could potentially be asked, well, where would the, For example, where would the sporadic form arise? And the sporadic form arises more commonly in the abdomen, the African form more commonly arises in the jaw. Very important to know about these tumors that they're driven by translocations of C-MYC. C-MYC is present on chromosome eight. The most common translocation is an 8/14 translocation. In this particular case, C-MYC gets translocated to the immunoglobulin heavy chain locus on 14, and that results in overexpression of C-MYC. Now, CMYC is a nuclear regulator that activates multiple genes necessary for promotion of growth. And so when you have a lot of CMYC, you're going to have a lot of growth, and that is then going to drive production of the tumor. So most important to recognize that it's CMYC that drives production of this tumor. This tumor is so rapidly growing, it has a very high mitotic rate, and that rapid growth results in a starry sky appearance on histology. Let me explain that a little further. Here's the histology, and these cells, all of these circular blue cells, these represent the tumor. And so they call this the sky. The blue represents the sky. And then here, some of the cells are

turning over so quickly that they're actually dying, and these are macrophages, i.e., tingible body macrophages, that are present within the tumor and are eating away the cells as they die. They're growing so fast that they die. So that is the star. So the blue represents the sky, and this white area represents the star. And the classic, again, description is a starry sky appearance. So that takes care of the intermediate-sized lymphoma of B-cells. Now we come to the lymphoma in which there are large B cells, and this is called diffuse large B cell lymphoma. Diffuse, it doesn't produce follicles, it doesn't produce a margin, it doesn't produce a mantle, it grows diffusely. Large, the cells are large. B, the cells are B cells, and hence, they're CD20 positive. And of course, this is a lymphoma. So that's the summary of diffuse large B-cell lymphoma. It's the most common form of non-Hodgkin's lymphoma, and it's clinically very aggressive. It would make sense that it's clinically aggressive because it's not well differentiated. It doesn't look like normal structure. It doesn't have small B-cells. And so because it's poorly differentiated, we would expect it to be clinically aggressive. Now, diffuse large B-cell lymphoma can arise sporadically, or it can arise from transformation of follicular lymphoma. And I've already explained that in detail when we talked about follicular lymphoma. This tumor presents in late adulthood, usually as an enlarging lymph node or as an extranodal mass. Once you've understood non-Hodgkin's lymphoma, you can now then move to Hodgkin's lymphoma. And Hodgkin's lymphoma is not too difficult to understand. Basically, the notion in Hodgkin's lymphoma is that instead of having a whole mass of malignant cells, so for example, in a non-Hodgkin's lymphoma, the entire mass is composed of malignant cells, Well, in a Hodgkin's lymphoma, you don't have a big mass of malignant cells. Instead, you have rare neoplastic cells called Reed-Sternberg cells, and those Reed-Sternberg cells secrete cytokines that draw in other inflammatory cells, which then results in production of a mass. And so the key hallmark of a Hodgkin's lymphoma is a Reed-Sternberg cell. The Reed-Sternberg cell is a large B cell, classically with a multi-lobed nucleus and a prominent nucleolus. And often, and usually, most commonly, they are CD15 and CD30 positive. Notice that there is no CD20 expression here, although it's clearly a B cell. They are CD15 and CD30 positive. Now here's a picture of a Reed-Sternberg cell. I put one here in the center. This is a large cell. It has a multi-lobed nucleus. Here's one lobe, here's another lobe. We can clearly see this nucleolus, this darker purple area, and this kind of gives the appearance of owl eyes, and so it's also called an owl-eyed nucleus. So that's a Reed-Sternberg cell. Now notice that there's one Reed-Sternberg cell, but that there are then hundreds of other cells that are sort of joining the party. And those are reactive cells, but it's the Reed-Sternberg cell along with all of the reactive cells that's actually producing the mass, and so that is going to be classic for a Hodgkin's lymphoma. Now, in the Reed-Sternberg cells, they secrete cytokines. And when they secrete cytokines, that's going to result in a few things. Number one, a subset of these patients can present with B symptoms. B symptoms are like fever,

chills, and night sweats. And it's actually due to the cytokines being produced by the Reed-Sternberg cells. In addition, those cytokines are going to attract lymphocytes, plasma cells, macrophages, and eosinophils, which are then going to cause the actual mass to form. So the bulk of the tumor, the bulk of the mass in this disorder, is actually reactive cells. Also, fibrosis may be present, and you'll see why that's important in a moment. As I've already stated, the reactive inflammatory cells make up the bulk of the tumor, and they also form the basis for classification. So that the subtypes of Hodgkin's lymphoma include, number one, nodular sclerosis. This is the most common, and I'll explain it in more detail in a moment. There's a lymphocyte-rich variant, where there's a ton of lymphocytes that are associated with the Reed-Sternberg cell. There's a mixed cellularity variant, where there's mixed cell type that's associated with the Reed-Sternberg cell, especially eosinophils. And there's a lymphocyte-depleted variant, which actually has minimal lymphocytes associated with the Reed-Sternberg cell. The most common subtype, as I've already explained, is nodular sclerosis subtype. The classic presentation in nodular sclerosis subtype is that the patient has an enlarging neck lymph node or an enlarging mediastinal lymph node, and is usually a young female. biopsy is then performed, and the biopsy reveals a lymph node that is divided by broad bands of fibrosis. And this is the classic histology on low power. This is a picture of a lymph node, and you can see that there's these broad pink bands of fibrosis, as I'm outlining here, and these broad pink bands of fibrosis are cutting the lymph node into these nodules. And so that's nodular sclerosis subtype, remember that the word sclerosis just means hard, and it's hard because it's got all this fibrotic tissue within it. At the same time, one of the classic findings on histology is that the Reed-Sternberg cells in this particular subtype often sit in these big open spaces. So notice that we've got this big open space with a Reed-Sternberg cell in it, and another big open space with a Reed-Sternberg cell in it. And so these spaces look like lakes, and so they're given the name lacunar cells. And lacunar cells are a special kind of Reed-Sternberg cell that is present in nodular subtype. The other subtypes are not necessarily super common, and it's often difficult to remember them, but I'll just tell you a few important principles. Number one, the lymphocyte rich has the best prognosis. Number two, the lymphocyte depleted has the worst prognosis. So the more the lymphocytes, the better, the less the lymphocytes, the worse. And the lymphocyte depleted happens to be seen in the elderly and in HIV positive individuals. One final principle is that in the mixed cellularity variant, you see abundant eosinophils. And very important to know, for examination purposes, that those abundant eosinophils are being called in by IL-5. So the Reed-Sternberg cell is overproducing cytokines, for example, IL-5, and it's IL-5 that draws in the eosinophils. In these patients, there may also be an eosinophilia, where they have extra eosinophils in the blood, and that's also due to increased IL-5 production by the Reed-Sternberg cells. And

that closes off our discussion of lymphoma. Not too difficult if you just keep sort of the big picture in mind as you go through them. Our next discussion will be plasma cell disorders.